

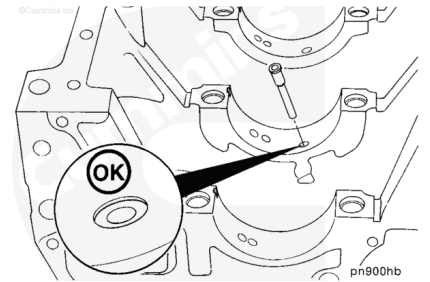
Trainings ▾

Initial Check

For engines equipped with drillings for saddle jet piston cooling nozzles/plugs, verify that:

- engines originally equipped with saddle jet piston cooling nozzles, the piston cooling nozzles are installed and free from debris/damage
- engines equipped with J-jet piston cooling nozzles, the saddle jet piston cooling nozzle drillings are plugged.

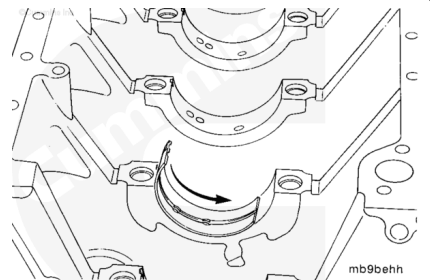
Note : Some engine do are **not** contain drillings for saddle jet piston cooling nozzles/plugs.



Install

Make sure the backsides of the bearings are clean and free of debris before installing the upper main bearings into the block.

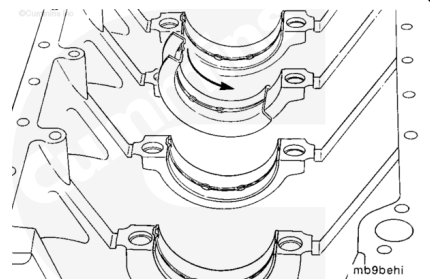
Make sure to align the tangs of the bearings with tangs on the main bearing block saddles.



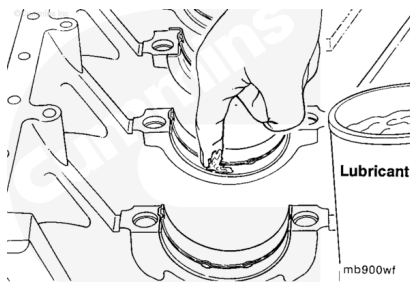
Install the upper crankshaft thrust bearing.

4 cylinder engines - The number 4 main bearing position.

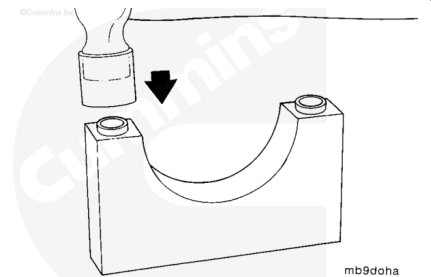
6 cylinder engines - The number 6 main bearing position.



Apply a coat of assembly lube, Part Number 3163087, to the crankshaft side of the main bearings and thrust bearing surfaces.

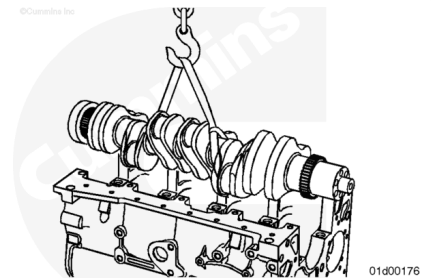


Check the main bearing caps to make sure the ring dowels are installed.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Carefully install the crankshaft to avoid damage to the crankshaft main bearings, especially the thrust/main bearing journals. Engine life will be shortened if damage to the crankshaft occurs.

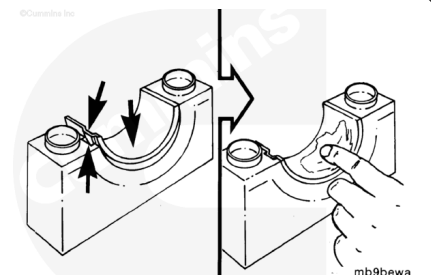
Install the crankshaft.

Note : When setting the crankshaft into the block, make sure the front crankshaft gear meshes with the lubricating oil pump gear (if not previously removed).

⚠ CAUTION ⚠

Do not lubricate the back side of the bearing that contacts the main bearing cap.

Make sure the backsides of the bearings are clean and free of debris before installing the lower main bearings into the main bearing caps.



Make sure to align the tangs of the bearings with tangs on the main bearing caps.

Note : Some engines use a thrust bearing for the upper and lower main bearing (360 degree), while other engines only have a thrust bearing for the upper main bearing (180 degree). Always replace like for like.

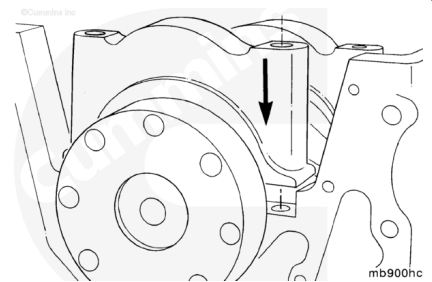
If equipped, install the lower crankshaft thrust bearing.

4 cylinder engines - The number 4 main bearing position.

6 cylinder engines - The number 6 main bearing position.

Apply a coat of assembly lube, Part Number 3163087, to the crankshaft side of the main bearings and thrust bearing surfaces.

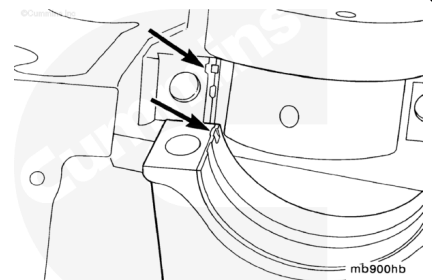
Make sure the main bear cap surfaces between the main bearing cap and block are clean and free of debris.



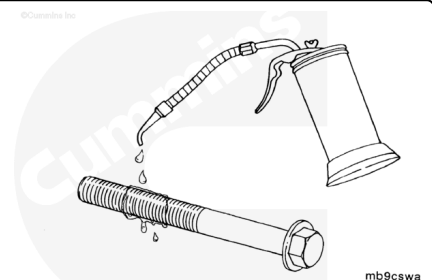
The main bearing caps are/were numbered during the removal process for their location. Number 1 starts with the front of the block.

Note : The caps must be installed so the numbers on the caps match the bearing saddle in the block. The lock tangs in the main bearing saddle and bearing cap must be on the same side.

Install the main bearing caps. Make sure to align the ring dowels on the main bearing cap with the corresponding drillings in the cylinder block.



Lubricate the main bearing capscrew threads and underside of the head with clean engine oil.

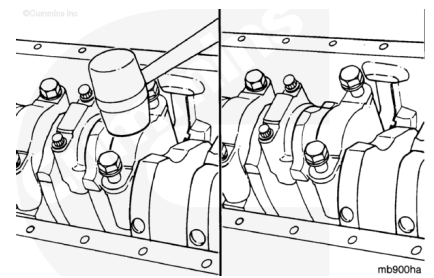


Gently tap the main bearing cap into position with a plastic or rubber mallet.

When seated, install the main bearing capscrews and tighten.

Torque Value: 50 n•m [37 ft-lb]

Do **not** tighten to the final torque value at this time. Final torque should be applied after all main bearing caps are installed.



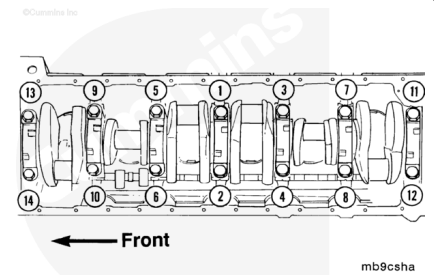
Note : The sequence to the right is for a 6 cylinder engine. For a 4 cylinder engine, use the same sequence for the 5 main bearing caps.

Tighten the capscrews evenly and in sequence. Perform each step to all capscrews before performing the next step.

B3.9, B4.5, and B5.9 Engines:

Torque Value:

1. 60 n•m [44 ft-lb]
2. 90 n•m [66 ft-lb]
3. Turn all capscrews through 90 degrees.



⚠ CAUTION ⚠

For B4.5 RGT engines, there is a different torque procedure for new and previously installed main bearing capscrews. Failure to use the correct torque value can result in engine damage.

B4.5 RGT Engines:

Torque Value:

Previously Installed Main Bearing Capscrews

1. 60 n•m [44 ft-lb]
2. 80 n•m [59 ft-lb]
3. Rotate 90 degrees.

Torque Value:

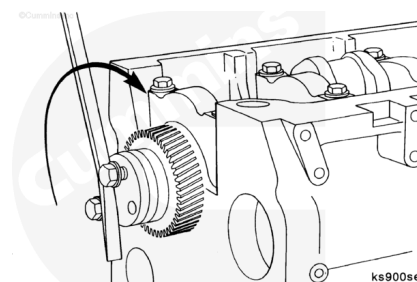
New Main Bearing Capscrews

1. 120 n•m [89 ft-lb]
2. Loosen completely.
3. 60 n•m [44 ft-lb]
4. 85 n•m [63 ft-lb]
5. Rotate 120 degrees.

The crankshaft **must** rotate freely after installing the main bearing caps.

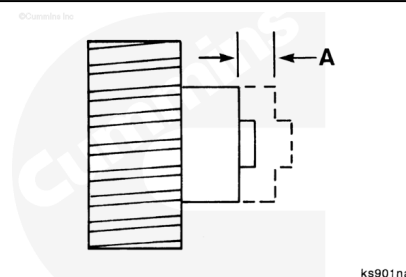
If the crankshaft does **not** rotate freely:

1. Check if the crankshaft is contacting one of the connecting rods
2. Check if the correct main bearing caps were installed correctly
3. Check if the main bearing cap ring dowels or mounting surfaces were damaged during installation
4. Check if the correct main bearings were installed.



Note : The dimensions of the thrust bearing and crankshaft journal determine end play.

Measure the crankshaft end play with a dial indicator assembly, Part Number 3824564 and magnetic base, Part Number 3377399.



Crankshaft End Play		
mm		in
0.102	MIN	0.004
0.432	MAX	0.017

If the crankshaft end play is not within specification:

1. If the crankshaft end play is below specification, check if there are any obstructions limiting the crankshaft's travel (lubricating oil pump, connecting rod, etc.)
2. If the crankshaft end play is above specification, inspect the crankshaft thrust bearing surface. Also check if the correct thrust bearing(s) were installed.

Note : Oversize thrust bearings are available if the end play is **not** within specifications. Oversize thrust bearings of 0.25 to 0.51 mm [0.010 to 0.020 in] are available.